



COVID-19 Study - Interpersonal Emotion Regulation Flexibility, Daily Diary, Negative and Positive Affect - Full

Public registration



Overview

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Study Information



Hypotheses

Research Questions:

- Emotion Regulation Flexibility as defined by Bonanno & Burton (2013) Context sensitivity
- Does the controllability of a negative event moderate the association of interpersonal emotion regulation strategy and affect?
- Repertoire Range
- Is repertoire range associated with affect?
- Responsiveness to Feedback
- When there is a change in interpersonal emotion regulation strategy use from the day prior (from t-1 to t), is the association between change in interpersonal emotion regulation strategy and affect (at t) moderated by affect levels the day before (t-1)?
- Emotion Regulation Flexibility as defined by Aldao, Sheppes, & Gross (2015)
4. Regressing emotion regulation strategy variability (at t) on environmental change is considered emotion regulation flexibility. Does this emotion regulation flexibility predict affect?

Hypotheses:

- Emotion Regulation Flexibility as defined by Bonanno & Burton (2013) Context sensitivity
- 1a. Perceived controllability of negative events will moderate the association between the use of co-problem-solving and negative affect, whereby at lower perceived controllability of a negative event, higher levels of co-problem solving will be associated with higher negative affect. Additionally, we predict that at higher perceived controllability of a negative event, higher levels of co-problem solving will be associated with lower negative affect.

Perceived controllability of negative events will moderate the association between the use of co-problem-solving and positive affect, whereby at higher perceived controllability of a negative event, higher levels of co-problem-solving will be associated with higher positive affect. Additionally, we predict that at lower perceived controllability of a negative event, higher levels of co-problem solving will be associated with lower positive affect.

- 1b. Perceived controllability of negative events will moderate the association between the use of co-reappraisal and negative affect, whereby at higher perceived controllability of a negative event, higher levels of co-reappraisal will be associated with higher negative affect. Additionally, we predict that at lower perceived controllability of a negative event, higher levels of co-reappraisal will be associated with lower negative affect.

Perceived controllability of negative events will moderate the association between the use of co-reappraisal and positive affect, whereby at higher perceived controllability of a negative event, higher levels of co-reappraisal will be associated with lower positive affect. Also, we predict that at lower perceived controllability of a negative event, higher levels of co-reappraisal will be associated with higher positive affect.

- 1c. Perceived controllability of negative events will moderate the association between the use of co-rumination and negative affect, whereby at lower perceived controllability of a negative event, higher levels of co-rumination will be associated with higher negative affect. Also, we predict that at higher perceived controllability of a negative event, higher levels of rumination will be associated with lower negative affect.

Perceived controllability of negative events will moderate the association between the use of co-rumination and positive affect, whereby at lower perceived controllability of a negative event, higher levels of co-rumination will be associated with lower positive affect. Also, we predict that at higher perceived controllability of a negative event, higher levels of co-rumination will be associated with higher positive affect.

- 1d. Perceived controllability of negative events will moderate the association between the use of co-distraction and negative affect, whereby at higher perceived controllability of a negative event, higher levels of co-distraction will be associated with higher negative affect. We also predict that at lower perceived controllability of a negative event, higher levels of co-distraction will be associated with lower negative affect.

Perceived controllability of negative events will moderate the association between the use of co-distraction and positive affect, whereby at higher perceived controllability of a negative event, higher levels of co-distraction will be associated with lower positive affect. Additionally, at lower perceived controllability of a negative event, higher levels of co-distraction will be associated with higher positive affect.

- Repertoire Range
2. Higher variability in interpersonal emotion regulation strategy use (categorical and temporal variability) will predict lower negative affect and higher positive affect, compared to low variability in interpersonal emotion regulation strategy use.
- Responsiveness to Feedback
3. When there is a change in interpersonal emotion regulation strategy use (from t-1 to t), the association between change in interpersonal emotion regulation strategy and negative affect (at t) will be moderated by negative affect levels the day before (at t-1). The change in interpersonal emotion regulation strategy use (from t-1 to t) will be associated with less negative affect (at t) at higher levels of negative affect the day before (at t-1).

When there is a change in interpersonal emotion regulation strategy use (from t-1 to t), the association between change in interpersonal emotion regulation strategy and positive affect (at t) will be moderated by positive affect levels the day before (at t-1). The change in interpersonal emotion regulation strategy use (from t-1 to t) will be associated with greater positive affect (at t) at lower levels of positive affect the day before (at t-1).

- Emotion Regulation Flexibility as defined by Aldao, Sheppes, & Gross (2015)
4. Regressing emotion regulation strategy variability (at t) on environmental change (indexed by the number of events, a change in event type from t-1 to t, and the change in event stressfulness from t-1 to t) is considered emotion regulation flexibility. We predict that higher emotion regulation flexibility will predict lower negative affect and higher positive affect, whereby lower emotion regulation flexibility will predict higher negative affect and lower positive affect.

Design Plan

Study type

Other

Blinding

No blinding is involved in this study.

Is there any additional blinding in this study?

No response

Study design

This study uses an online survey longitudinal method. Participants completed a baseline questionnaire and 14-day daily diaries.

No files selected

Randomization

No response

Sampling Plan

Existing Data

Registration prior to analysis of the data

Explanation of existing data

No response

Data collection procedures

Participants were recruited for the COVID-19 and Wellbeing Study (baseline survey and daily diaries) from April 20th-July 20th 2020, from Canada and the United States. This study is an online long-term longitudinal study investigating the effects of COVID-19 on well-being. Participants consented to not receive remuneration for this study. Participants were required to be fluent in English and over age 18 to participate in the study.

No files selected

Sample size

Participants were recruited for the COVID-19 and Wellbeing Study (baseline survey and daily diaries) from April 20th-July 20th 2020, from Canada and the United States. This study is an online long-term longitudinal study investigating the effects of COVID-19 on well-being. Participants consented to not receive remuneration for this study. Participants were required to be fluent in English and over age 18 to participate in the study. If participants reported that they were not fluent in English and/or not age 18 or above, they will be excluded from analyses. Participants will be included in this current pre-registered study if they passed all three attention checks. Only participants that completed more than 1 daily diary will be included in analyses.

Sample size rationale

No response

Stopping rule

No response

Variables

Manipulated variables

No response

No files selected

Measured variables

Dependent variables:

Affect. Items derived from The Positive and Negative Affect Schedule (PANAS; Watson, Clark, & Tellegen, 1988) were used to assess affect before bedtime, daily for 14-days. The negative affect items assessed: distressed, upset, guilty, scared, hostile, irritable, ashamed, nervous, jittery, afraid. The positive affect items assessed: interested, excited, strong, enthusiastic, proud, alert, inspired, determined, attentive, active. Participants indicated to what degree they felt each emotion that day on a scale from 1 (Very slightly or not at all) to 5 (Extremely). The items were summed for a score of negative and positive affect, respectively.

Independent variables:

Event. Participants were asked to report on the most significant event that occurred for them that day. They were provided a list of 11 types of events to report (close friendship, social life, romantic relationship, family relationship, education, work, finances, health of self, health of family, health of friends, recreation). Participants also had the choice of "other" where they could indicate the type of event in an open-ended response format.

Event Pleasantness. If participants reported an event, they were asked to report how pleasant the event was on a scale from 1 (Very unpleasant) to 5 (Very pleasant). Negative events will be characterized as ratings of 1 and 2 on this pleasantness scale.

Event Controllability. If participants reported an event, they were asked "How controllable was the event?" on a scale from 1 (Very uncontrollable) to 5 (Very controllable).

Event Stressfulness. If participants reported an event, they were asked "How stressful was the event?" on a scale from 1 (Not at all stressful) to 5 (Extremely stressful).

Emotion Regulation. Participants were still asked to report on the emotion regulation strategies used that day. Participants rated their use of each interpersonal emotion regulation strategy (i.e., co-problem-solving, co-reappraisal, co-rumination, co-distraction) on a scale from 1 (Not at all) to 5 (Very much so).

No files selected

Indices

No response

No files selected

Analysis Plan

Statistical models

Hypotheses:

Multilevel Modelling (MLM) and regression analyses will be conducted for our analyses. Regression analyses will be used for research questions not requiring a nested analytic approach, whereas MLM will be used for research questions requiring nested levels. For our MLM analyses, we will use a two-level model with days (Level 1) nested within persons (Level 2). The specific analyses corresponding to each of the 4 research questions are indicated below:

- Five MLM models will be conducted for each interpersonal emotion regulation strategy, at Level 1, negative and positive affect at time t will be predicted by the interpersonal emotion regulation strategy (e.g., co-problem-solving) and controllability at time t, and the interaction of these variables. The predictors will be person-centred. The predictor (interpersonal emotion regulation strategy) will also be grand mean centred at Level 2.

- We will assess both categorical and temporal variability. The analytic plan here and the terminology are consistent with past research (Blanke et al., 2020; Eldesouky & English, 2018). In terms of categorical variability, interpersonal emotion regulation strategy use (or lack thereof) will be dichotomized each day. For instance, if participants reported they did not use the strategy (1 Not at all), this will be coded as 0, and for strategies rated 2 and above on the 7-point scale, this will be coded as 1 (the strategy was implemented). These codes will be summed. A higher total will equate to a broader range of strategies used each day. Using MLM, at Level 1, this categorical variability at t will predict negative and positive affect at t. Categorical variability at Level 1 will be person-centred. The predictor (categorical variability) will also be grand mean centred at Level 2.

With regards to temporal variability, we will first compute the standard deviations (SDs) for the use of each interpersonal emotion regulation strategy (within-strategy variability) for each participant. The within-strategy variability will be obtained by taking the SD of each strategy used across all time points (more than one assessment is needed to compute within-strategy variability). We will then average the SDs of each strategy for an overall average of within-strategy variability for each participant. For affect (outcome variable) we will take the average of affect reported by each participant across all time points (negative and positive affect separately). We will conduct a regression analysis to test within-strategy variability predicting negative and positive affect. Consistent with Blanke et al., (2020), we will control for the mean of emotion regulation strategy use in our analyses.

The between-strategy variability will be obtained by computing the SD of strategy use at each time point (as multiple strategies can be endorsed at each time point). We will control for the mean of emotion regulation strategy use at each time point in our analysis. We will conduct an MLM model to test between-strategy variability at t, predicting negative and positive affect at t. The predictor (between-strategy variability) and control variable (mean of emotion regulation strategy use) will be person mean centred at Level 1. The predictor (between-strategy variability) will also be grand mean centred at Level 2.

- Time-lagged variables will be created as t-1 and t. A change variable will be computed for each interpersonal emotion regulation strategy by subtracting the amount the strategy was used at t from t-1. We will take the absolute value and average this value across the strategies. At Level 1, affect (negative and positive for their respective models) at t-1, the emotion regulation change variable, and an interaction variable will be defined as affect (negative and positive for their respective models) at t-1*emotion regulation change, in predicting negative and positive affect at t. These Level 1 predictors will be person-centred. The predictor (emotion regulation change) will also be grand mean centred at Level 2.

- This statistical analysis was derived from suggestions from Aldao, Sheppes, & Gross (2015). For within-strategy variability, a regression model will be conducted that examines the use of interpersonal emotion regulation over event type variability (the number of the types of events reported). We will save the residuals from this model to indicate within-strategy emotion regulation flexibility. Positive residuals indicate higher emotion regulation variability compared to event variability which indicates higher emotion regulation flexibility relative to the rest of the sample. Negative residuals indicate lower emotion regulation variability compared to event variability which indicates lower emotion regulation flexibility in relation to the rest of the sample. We will use the saved residuals (defined as within-strategy emotion regulation flexibility) to predict negative and positive affect (average negative and positive affect reported by each participant across all time points). Results will indicate whether individuals with higher or lower within-strategy emotion regulation flexibility predicts negative and positive affect.

For between-strategy variability, as Aldao, Sheppes, & Gross (2015) suggested, we will first lag environmental change in a multilevel model to predict between-strategy variability. Change in event type from t-1 to t will be coded as 0 (no change) and 1 (change), at Level 1. This will predict between-strategy variability at t. The residuals of this model will be saved. For a second MLM model, the residuals will predict negative and positive affect at t. This model will also have between-strategy variability grand mean centered at Level 2.

This model will also be conducted with the predictor as the change in the stressfulness of the event from t-1 to t, at Level 1. This will predict between-strategy variability at t. The residuals of this model will be saved. For a second MLM model, the residuals will predict negative and positive affect at t. This model will also have between-strategy variability grand mean centered at Level 2.

No files selected

Transformations

No response

Inference criteria

No response

Data exclusion

Outliers will not be removed given that outliers from our study variables present meaningful information. This current pre-registration is for the analysis of the daily diary portion of the long-term longitudinal study. If participants did not complete any daily diaries, they will not be included in the analyses. Daily diary entries completed from 5pm-6am in the participants timezone will be included, while entries completed outside of this window will be excluded. If participants completed duplicate diaries (two in one day), only their first entry will be used. Daily diary entries will be included if they were completed during the 14-days period. The day and time diaries were completed are based on the timezone participants' reported in the baseline survey; thus, any participants that did not report their timezone in the baseline survey cannot be included in analyses. For a subset of analyses, only participants that reported experiencing a negative event or those that reported experiencing events in general (negative and positive) will be included, for Hypothesis 1 and 4, respectively. Only participants that reported interacting with people and thus, using interpersonal strategies, will be included in all analyses. Only participants that completed more than 1 daily diary will be included in analyses.

Missing data

No response

Exploratory analysis

No response

Other

Other

No response

Contributors

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Description

This study is from the COVID-19 and Wellbeing Study.

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OSF Preregistration

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Psychology

Social and Behavioral Sciences

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